

Manual Website Setup Using Vagrant

```
MINGW64/f/vagrant-vms/cafe
USER@DESKTOP-V51CFFH MINGW64 ~ (main)
$ cd /f/

USER@DESKTOP-V51CFFH MINGW64 /f
$ mkdir vagrant-vms

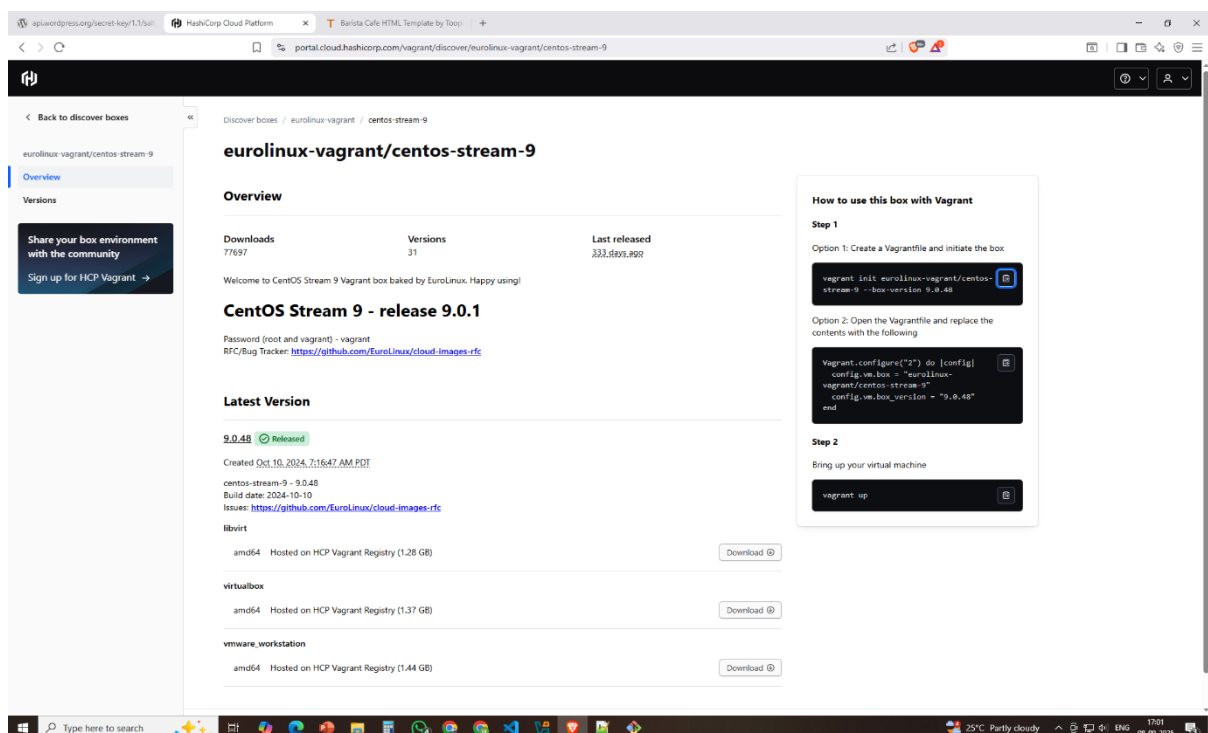
USER@DESKTOP-V51CFFH MINGW64 /f
$ cd vagrant-vms/

USER@DESKTOP-V51CFFH MINGW64 /f/vagrant-vms
$ mkdir cafe

USER@DESKTOP-V51CFFH MINGW64 /f/vagrant-vms
$ cd cafe

USER@DESKTOP-V51CFFH MINGW64 /f/vagrant-vms/cafe
$ ls
```

Set up the vagrant-vms directory and created a website folder inside it for hosting



Used **vagrant init** to activate a Vagrant Cloud box and set up the Vagrantfile

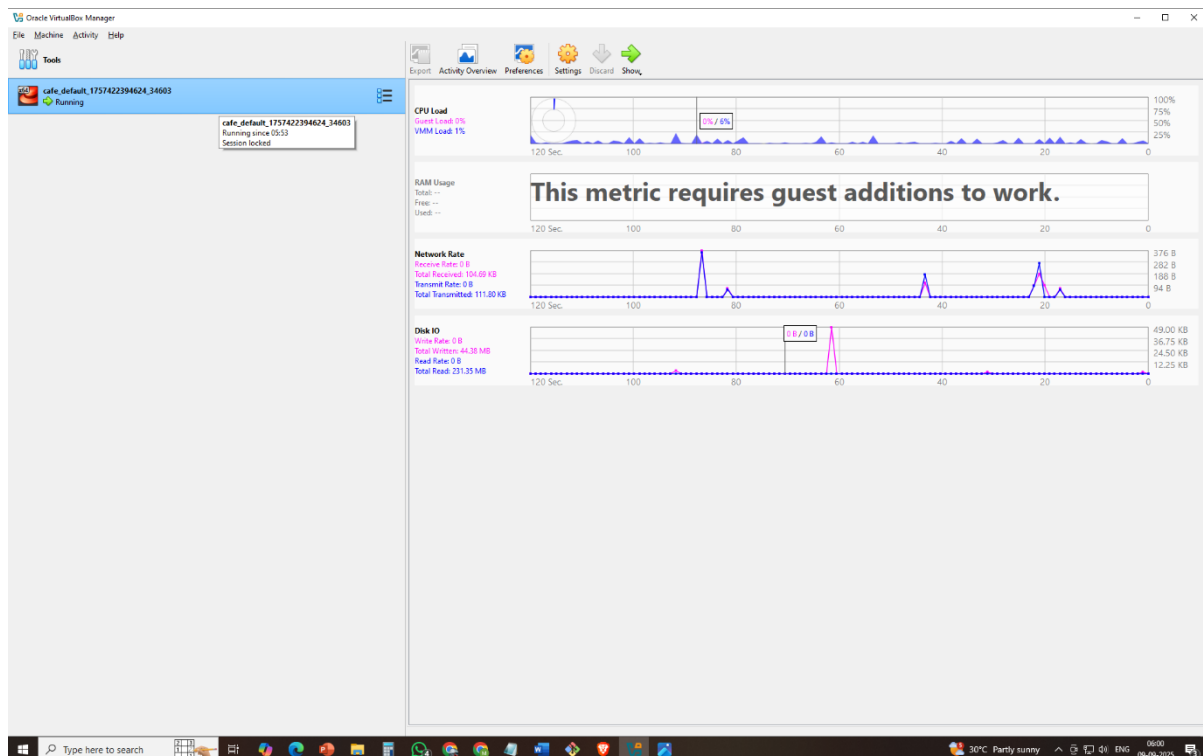
```
MINGW64 /f/vagrant-vms/cafe
USER@DESKTOP-V51CFFH MINGW64 /f/vagrant-vms/cafe
$ vagrant init eurolinux-vagrant/centos-stream-9 --box-version 9.0.48
A Vagrantfile has been placed in this directory. You are now
ready to 'vagrant up' your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
'vagrantup.com' for more information on using Vagrant.

USER@DESKTOP-V51CFFH MINGW64 /f/vagrant-vms/cafe
$ ls
Vagrantfile

USER@DESKTOP-V51CFFH MINGW64 /f/vagrant-vms/cafe
$ vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Importing base box 'eurolinux-vagrant/centos-stream-9'...
==> default: Matching MAC address for NAT networking...
==> default: Checking if box 'eurolinux-vagrant/centos-stream-9' version '9.0.48' is up to date...
==> default: Setting the name of the VM: cafe_default_1757422394624_34603
==> default: Clearing any previously set network interfaces...
==> default: Preparing network interfaces based on configuration...
==> default: Adapter 1: nat
==> default: Forwarding ports...
==> default: 22 (guest) => 2222 (host) (adapter 1)
==> default: Booting VM...
==> default: Waiting for machine to boot. This may take a few minutes...
==> default: SSH address: 127.0.0.1:2222
==> default: SSH username: vagrant
==> default: SSH auth method: private key
==> default:
==> default: Vagrant insecure key detected. Vagrant will automatically replace
==> default: this with a newly generated keypair for better security.
==> default:
==> default: Inserting generated public key within guest...
==> default: Removing insecure key from the guest if it's present...
==> default: Key inserted! Disconnecting and reconnecting using new SSH key...
==> default: Machine booted and ready!
==> default: Checking for guest additions in VM...
==> default: No guest additions were detected on the base box for this VM! Guest
==> default: additions are required for forwarded ports, shared folders, host only
==> default: networking, and more. If SSH fails on this machine, please install
==> default: the guest additions and repackaging the box to continue.
==> default:
==> default: This is not an error message; everything may continue to work properly,
==> default: in which case you may ignore this message.
==> default: Rsyncing folder: /cygdrive/f/vagrant-vms/cafe/ => /vagrant

USER@DESKTOP-V51CFFH MINGW64 /f/vagrant-vms/cafe
$
```

This shows the Vagrantfile in the directory and the **vagrant up** command used to start the CentOS machine



Verified in VirtualBox that the machine is up and running

```
root@cafesite:~  
  
This box was generated by EuroLinux  
Any suggestions are welcome at https://github.com/EuroLinux/cloud-images-rfc/  
  
Happy using.  
To delete this message use:  
echo '' > /etc/motd  
[vagrant@vbox ~]$ sudo -i  
[root@vbox ~]# vim /etc/hostname  
[root@vbox ~]# exit  
logout  
[vagrant@vbox ~]$ exit  
logout  
  
USER@DESKTOP-V51CFFH MINGW64 /f/vagrant-vm/cafesite  
$ vagrant ssh  
This box was generated by EuroLinux  
Any suggestions are welcome at https://github.com/EuroLinux/cloud-images-rfc/  
  
Happy using.  
To delete this message use:  
echo '' > /etc/motd  
Last login: Tue Sep 9 06:02:00 2025 from 10.0.2.2  
[vagrant@vbox ~]$ sudo -i
```

Accessed the Vagrant machine with **vagrant ssh**, switched to root, and changed the hostname using **vim /etc/hostname**. This step is optional.

Used Git Bash to **vagrant ssh** into the machine, logged in as root, and modified the hostname. (Not a required step.)

```
root@cafesite:~  
[root@cafesite ~]# yum  
[root@cafesite ~]# yum install httpd wget unzip zip -y  
Last metadata expiration check: 0:02:30 ago on Tue 09 Sep 2025 06:15:30 AM UTC.  
Package wget-1.21.1-8.el9.x86_64 is already installed.  
Package unzip-6.0-59.el9.x86_64 is already installed.  
Package zip-3.0-35.el9.x86_64 is already installed.  
Dependencies resolved.  
  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
httpd	x86_64	2.4.62-7.el9	appstream	46 k
Installing dependencies:				
apr	x86_64	1.7.0-12.el9	appstream	123 k
apr-util	x86_64	1.6.1-23.el9	appstream	95 k
apr-util-bdb	x86_64	1.6.1-23.el9	appstream	13 k
centos-logos-httpd	noarch	90.8-3.el9	appstream	1.5 M
httpd-core	x86_64	2.4.62-7.el9	appstream	1.5 M
httpd-filesystem	noarch	2.4.62-7.el9	appstream	11 k
httpd-tools	x86_64	2.4.62-7.el9	appstream	80 k
Installing weak dependencies:				
apr-util-openssl	x86_64	1.6.1-23.el9	appstream	15 k
mod_http2	x86_64	2.0.26-5.el9	appstream	163 k
mod_lua	x86_64	2.4.62-7.el9	appstream	58 k

```
=====
```

Transaction Summary

Install 11 Packages

Total download size: 3.6 M
Installed size: 8.6 M
Downloading Packages:

Package	Download Size	Progress	Time Left
(1/11): apr-util-bdb-1.6.1-23.el9.x86_64.rpm	13 kB	28 kB/s	00:00
(2/11): apr-util-openssl-1.6.1-23.el9.x86_64.rpm	15 kB	158 kB/s	00:00
(3/11): apr-1.7.0-12.el9.x86_64.rpm	123 kB	212 kB/s	00:00
(4/11): httpd-2.4.62-7.el9.x86_64.rpm	46 kB	517 kB/s	00:00

Used **yum** to install Apache (httpd) along with **wget**, **unzip**, and **zip** utilities.

Installed Apache and supporting tools (**wget**, **unzip**, **zip**) with the **yum** package manager.

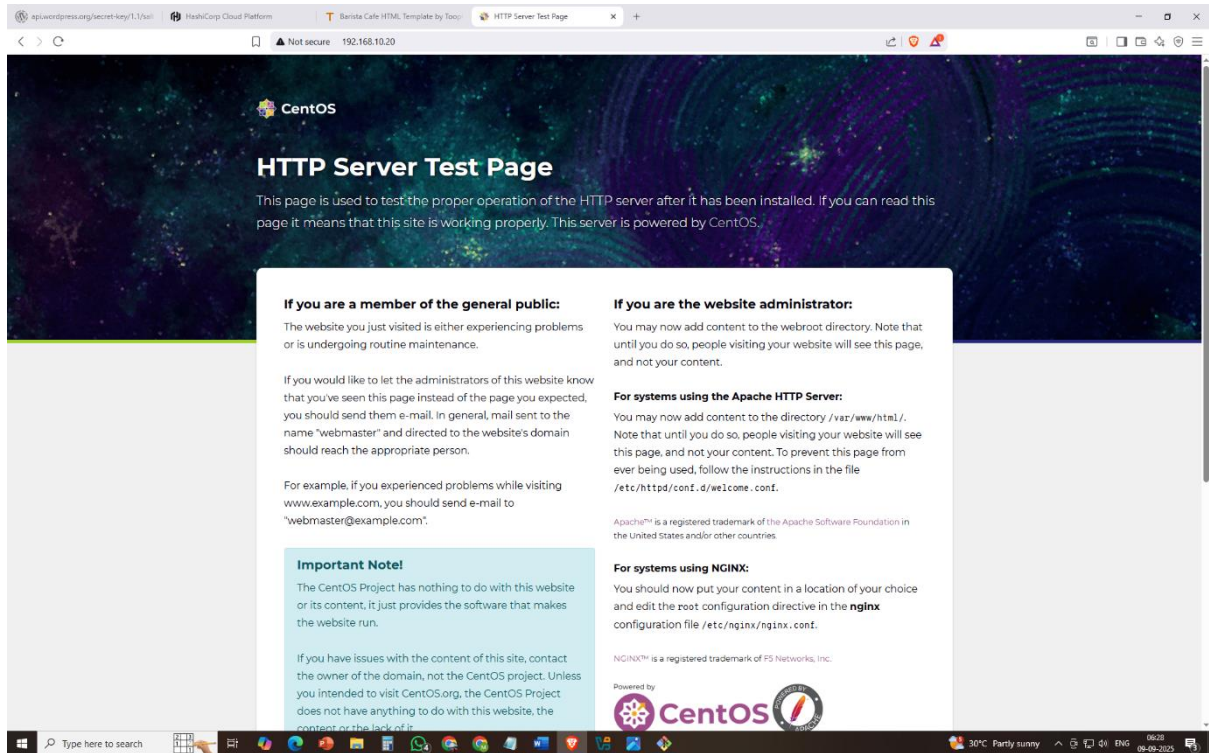
```
[root@cafesite ~]# systemctl start httpd
[root@cafesite ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@cafesite ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2025-09-09 06:20:12 UTC; 27s ago
     Docs: man:httpd.service(8)
  Main PID: 42305 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 11113)
    Memory: 33.6M (peak: 33.9M)
       CPU: 126ms
    CGroup: /system.slice/httpd.service
            └─42305 /usr/sbin/httpd -DFOREGROUND
              └─42306 /usr/sbin/httpd -DFOREGROUND
                └─42307 /usr/sbin/httpd -DFOREGROUND
                  └─42308 /usr/sbin/httpd -DFOREGROUND
                    └─42309 /usr/sbin/httpd -DFOREGROUND

Sep 09 06:20:11 cafesite systemd[1]: Starting The Apache HTTP Server...
Sep 09 06:20:11 cafesite httpd[42305]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, setting 'ServerName' to '::'.
Sep 09 06:20:12 cafesite systemd[1]: Started The Apache HTTP Server.
Sep 09 06:20:12 cafesite httpd[42305]: Server configured, listening on: port 80
[root@cafesite ~]#
```

Started the HTTP server using `systemctl start httpd`, enabled it to run on boot with `systemctl enable httpd`, and verified the service status using `systemctl status httpd`

```
[root@cafesite ~]# ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:e4:ec:85 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86315sec preferred_lft 86315sec
   inet6 fd17:625c:f037:2:a00:27ff:fee4:ec85/64 scope global dynamic noprefixroute
       valid_lft 86316sec preferred_lft 14316sec
   inet6 fe80::a00:27ff:fee4:ec85/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:3a:15:8a brd ff:ff:ff:ff:ff:ff
   inet 192.168.10.20/24 brd 192.168.10.255 scope global noprefixroute enp0s8
       valid_lft forever preferred_lft forever
   inet6 fe80::a00:27ff:fe3a:158a/64 scope link
       valid_lft forever preferred_lft forever
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:6d:e8:10 brd ff:ff:ff:ff:ff:ff
   inet 192.168.37.148/24 brd 192.168.37.255 scope global dynamic noprefixroute enp0s9
       valid_lft 3515sec preferred_lft 3515sec
   inet6 fe80::a00:27ff:fe6d:e810/64 scope link
       valid_lft forever preferred_lft forever
[root@cafesite ~]#
```

Checked the server's IP address using `ip addr show` and used it in the browser to test the default Apache web page.



Default web page was accessible. Test page httpd

```
root@cafesite- [root@cafesite tmp]# wget https://www.tooplate.com/zip-templates/2137_barista_cafe.zip
--2025-09-09 06:31:08-- https://www.tooplate.com/zip-templates/2137_barista_cafe.zip
Resolving www.tooplate.com (www.tooplate.com)... 72.52.176.250, 64:ff9b::4834:b0fa
Connecting to www.tooplate.com (www.tooplate.com) [72.52.176.250]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4679329 (4.5M) [application/zip]
Saving to: '2137_barista_cafe.zip'

2137_barista_cafe.zip 100%[=====] 4.46M 579KB/s in 9.2s

2025-09-09 06:31:19 (495 KB/s) - '2137_barista_cafe.zip' saved [4679329/4679329]

[root@cafesite tmp]# ls
2137_barista_cafe.zip
systemd-private-75900c5ccccf4315ad39d236bcf692f1-chrnyd.service-zMYTfy
systemd-private-75900c5ccccf4315ad39d236bcf692f1-dbus-broker.service-2NM84T
systemd-private-75900c5ccccf4315ad39d236bcf692f1-httpd.service-cvSJaO
systemd-private-75900c5ccccf4315ad39d236bcf692f1-irqbalance.service-sLO3XC
systemd-private-75900c5ccccf4315ad39d236bcf692f1-kdump.service-iUZ5h1
systemd-private-75900c5ccccf4315ad39d236bcf692f1-systemd-logind.service-eCAHCQ
tmpczv3nk9y
[root@cafesite tmp]# unzip 2137_barista_cafe.zip
Archive: 2137_barista_cafe.zip
  creating: 2137_barista_cafe/
  creating: 2137_barista_cafe/css/
  inflating: 2137_barista_cafe/css/bootstrap-icons.css
  inflating: 2137_barista_cafe/css/bootstrap.min.css
  inflating: 2137_barista_cafe/css/tooplate-barista.css
  inflating: 2137_barista_cafe/css/vegas.min.css
  creating: 2137_barista_cafe/fonts/
  inflating: 2137_barista_cafe/fonts/bootstrap-icons.woff
  inflating: 2137_barista_cafe/fonts/bootstrap-icons.woff2
  creating: 2137_barista_cafe/images/
  inflating: 2137_barista_cafe/images/barman-with-fruits.jpg
  inflating: 2137_barista_cafe/images/coffee-beans.png
  inflating: 2137_barista_cafe/images/happy-loving-couple-bakers-drinking-coffee-looking-notebook.jpg
```

Downloaded a website template using **wget**, verified the file with **ls**, and extracted it using **unzip** **2137_barista_cafe.zip**.


```
[root@cafesite ~]# cd /tmp/
[root@cafesite tmp]# ls
2137_barista_cafe
2137_barista_cafe.zip
systemd-private-75900c5ccccf4315ad39d236bcf692f1-chrond.service-zMYTfy
systemd-private-75900c5ccccf4315ad39d236bcf692f1-dbus-broker.service-2NM84T
systemd-private-75900c5ccccf4315ad39d236bcf692f1-httpd.service-cVsJa0
systemd-private-75900c5ccccf4315ad39d236bcf692f1-irqbalance.service-sL03XC
systemd-private-75900c5ccccf4315ad39d236bcf692f1-kdump.service-iUZ5h1
systemd-private-75900c5ccccf4315ad39d236bcf692f1-systemd-logind.service-eCahCQ
tmpczv3nk9y
[root@cafesite tmp]# cd 2137_barista_cafe/
[root@cafesite 2137_barista_cafe]# ls
'ABOUT THIS TEMPLATE.txt'  css  fonts  images  index.html  js  reservation.html  videos
[root@cafesite 2137_barista_cafe]# cp -r* /var/www/html/
cp: invalid option -- '*'
Try 'cp --help' for more information.
[root@cafesite 2137_barista_cafe]# cp -r * /var/www/html/
[root@cafesite 2137_barista_cafe]# ls
'ABOUT THIS TEMPLATE.txt'  css  fonts  images  index.html  js  reservation.html  videos
[root@cafesite 2137_barista_cafe]# cd ..
[root@cafesite tmp]# ls /var/www/html/
'ABOUT THIS TEMPLATE.txt'  css  fonts  images  index.html  js  reservation.html  videos
[root@cafesite tmp]# cd
[root@cafesite ~]#
```

Transferred the website template into the Apache web root (**/var/www/html/**) and confirmed the files were in place."

Used **cp -r *** to copy all website files to **/var/www/html/** and verified the hosting directory

```
reservation.html
[root@cafesite 2137_barista_cafe]# cd ..
[root@cafesite tmp]# cd ..
Active: inactive (dead)
Docs: man:firewalld(1)
[root@cafesite ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2025-09-09 13:25:34 UTC; 6h left
   Docs: man:httpd.service(8)
  Main PID: 660 (httpd)
    Status: "Total requests: 3; Idle/Busy workers 100/0; Requests/sec: 6.98e-10; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 4638)
   Memory: 21.6M (peak: 21.9M)
      CPU: 2.142s
   CGroup: /system.slice/httpd.service
           └─660 /usr/sbin/httpd -DFOREGROUND
             └─711 /usr/sbin/httpd -DFOREGROUND
               └─712 /usr/sbin/httpd -DFOREGROUND
                 └─713 /usr/sbin/httpd -DFOREGROUND
                   └─714 /usr/sbin/httpd -DFOREGROUND

Sep 09 13:25:33 cafesite systemd[1]: Starting The Apache HTTP Server...
Sep 09 13:25:34 cafesite httpd[660]: AH00558: httpd: could not reliably determine the server's fully qualified
cafesite systemd[1]: Started The Apache HTTP Server.
Sep 09 13:25:34 cafesite httpd[660]: Server configured, listening on: port 80
[root@cafesite ~]# ls -l /va
vagrant/ var/
[root@cafesite ~]# ls -l /var/www/html/
total 56
-rw-r--r--. 1 root root 510 Sep 9 06:36 'ABOUT THIS TEMPLATE.txt'
drwxr-xr-x. 2 root root 107 Sep 9 06:36 css
drwxr-xr-x. 2 root root 63 Sep 9 06:36 fonts
drwxr-xr-x. 5 root root 4096 Sep 9 06:36 images
-rw-r--r--. 1 root root 34816 Sep 9 06:36 index.html
drwxr-xr-x. 2 root root 152 Sep 9 06:36 js
-rw-r--r--. 1 root root 11734 Sep 9 06:36 reservation.html
drwxr-xr-x. 2 root root 43 Sep 9 06:36 videos
[root@cafesite ~]#
```

```
[root@cafesite ~]# ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e4:ec:85 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 84625sec preferred_lft 84625sec
    inet6 fd17:625c:f037:2:a00:27ff:fee4:ec85/64 scope global dynamic noprefixroute
        valid_lft 86284sec preferred_lft 14284sec
    inet6 fe80::a00:27ff:fee4:ec85/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3a:15:8a brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.20/24 brd 192.168.10.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe3a:158a/64 scope link
        valid_lft forever preferred_lft forever
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6d:e8:10 brd ff:ff:ff:ff:ff:ff
    inet 192.168.37.148/24 brd 192.168.37.255 scope global dynamic noprefixroute enp0s9
        valid_lft 1825sec preferred_lft 1825sec
    inet6 fe80::a00:27ff:fe6d:e810/64 scope link
        valid_lft forever preferred_lft forever
[root@cafesite ~]#
```

